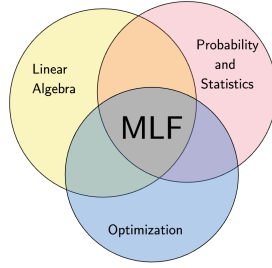


Week-6



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1. Matrix Decomposition

The idea is to decompose a matrix into the product of other matrices. For diagonalizable matrices A , we have matrices P, D such that $A = PDP^{-1}$. For orthogonally diagonalizable matrices A , we have $A = QDQ^T$. More recently, in the case of complex matrices, we saw that Hermitian matrices A are unitarily diagonalizable, namely, $A = UDU^*$.

$$\begin{array}{ll} A = PDP^{-1} & \text{any diagonalizable matrix} \\ A = QDQ^T & \text{real symmetric matrices} \\ A = UDU^* & \text{Hermitian matrices} \end{array}$$

While this is good, can we say something about any $m \times n$ matrix?

2. SVD

The singular value decomposition is a technique that is applicable for any matrix of shape $m \times n$. Formally, given any real matrix A of shape $m \times n$, there exist orthogonal matrices U, V and a diagonal-like matrix Σ such that:

$$A = U\Sigma V^T$$

For the complex case (or more general case), we have unitary matrices U and V such that:

$$A = U\Sigma V^*$$

The matrix Σ continues to remain real even for the complex case. Some more terminology

- The columns of U are called the left singular vectors.
- The columns of V are called the right singular vectors.
- The diagonal elements of Σ are called the singular values.

The shapes of the matrices:

$$\underbrace{A}_{m \times n} = \underbrace{U}_{m \times m} \underbrace{\Sigma}_{m \times n} \underbrace{V^T}_{n \times n}$$

- The columns of U form an orthonormal basis for $\mathbb{R}^m$.
- The columns of V form an orthonormal basis for $\mathbb{R}^n$.

Regarding the singular values:

- The singular values are all non-negative.
- They are represented in decreasing order on the diagonals.
- The number of singular values is equal to $\min(m, n)$. For example:

$$\Sigma = \begin{bmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \sigma_3 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \sigma_3 \end{bmatrix}$$

The SVD of A is not unique. However, the Σ matrix is going to be the same. We can also represent the SVD of A^T as:

$$A^T = V\Sigma^T U^T$$

3. $A^T A$ and AA^T

Let us assume that the SVD of A is given by $A = U\Sigma V^T$.

$$\begin{aligned}
A^T A &= (V \Sigma^T U^T) (U \Sigma V^T) \\
&= V \Sigma^T U^T U \Sigma V^T \\
&= V (\Sigma^T \Sigma) V^T
\end{aligned}$$

What we have is the eigen-decomposition of $A^T A$:

$$\underbrace{A^T A}_{n \times n} = \underbrace{V}_{n \times n} \underbrace{(\Sigma^T \Sigma)}_{n \times n} \underbrace{V^T}_{n \times n}$$

- The columns of V are the eigenvectors of $A^T A$
- The diagonal of $\Sigma^T \Sigma$ are the corresponding eigenvalues of $A^T A$.
The eigenvalues are the squares of the singular values of A .

For example, let $m = 4, n = 3$:

$$\Sigma = \begin{bmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \sigma_3 \end{bmatrix}$$

Now:

$$\Sigma^T \Sigma = \begin{bmatrix} \sigma_1^2 & & \\ & \sigma_2^2 & \\ & & \sigma_3^2 \end{bmatrix}$$

If we denote the eigenvalues of $A^T A$ as λ_i , then $\lambda_i = \sigma_i^2$. Similarly, we can repeat the process for AA^T .

$$AA^T = U(\Sigma \Sigma^T)U^T$$

Once again, the columns of U are the eigenvectors of AA^T and the squares of the singular values are the corresponding eigenvalues. Some properties that show the link between $A^T A$ and AA^T .

Property-1: The eigenvalues of $A^T A$ and AA^T are non-negative.

Property-2: $A^T A$ and AA^T have the same non-zero eigenvalues. They could have a different number of zero-eigenvalues and indeed will have a different number when $m \neq n$.

Property-3: $\text{rank}(A^T A) = \text{rank}(AA^T) = \text{rank}(A) = r$.

Property-4: There are a total of $\min(m, n)$ singular values, out of which, r are positive, where $r = \text{rank}(A)$. They are ordered as $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0 = \sigma_{r+1} = \sigma_{r+2} = \dots = \sigma_{\min(m,n)}$.

To summarize, the matrices involved are:

$$U = \begin{bmatrix} | & & | \\ u_1 & \cdots & u_m \\ | & & | \end{bmatrix}$$

$$V = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix}$$

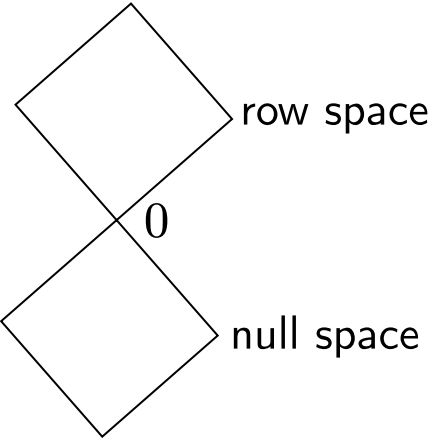
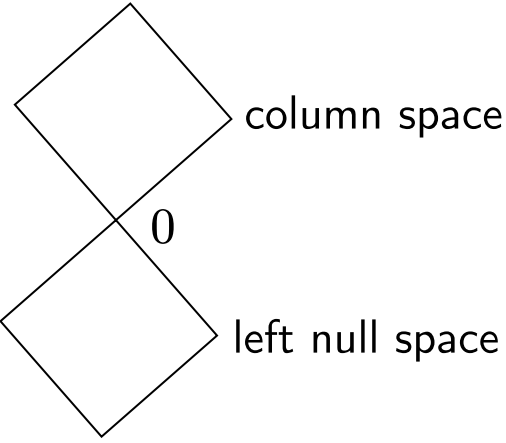
$$\Sigma = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

- (σ_i^2, u_i) is an eigenpair for AA^T for $1 \leq i \leq r$. The rest of the vectors starting from $u_{r+1}, \dots, u_m$ are eigenvectors with eigenvalue 0.
- (σ_i^2, v_i) is an eigenpair for $A^T A$ for $1 \leq i \leq r$. The rest of the vectors starting from $v_{r+1}, \dots, v_n$ are eigenvectors with eigenvalue 0.

For detailed proofs, refer to this [document](#).

4. Fundamental Theorem of Linear Algebra

This theorem connects the four fundamental subspaces of a matrix with its SVD. If you understand SVD thoroughly, then you understand linear algebra. Sort of. Let the SVD of $A = U\Sigma V^T$.

$\mathbb{R}^n$	$\mathbb{R}^m$
Row space This is the span of the first r right singular vectors. $\text{span}\{v_1, \cdots, v_r\}$	Column space This is the span of the first r left singular vectors. $\text{span}\{u_1, \cdots, u_r\}$
Null space This is the span of the last $n - r$ right singular vectors. $\text{span}\{v_{r+1}, \cdots, v_n\}$	Left null space This is the span of the last $m - r$ left singular vectors. $\text{span}\{u_{r+1}, \cdots, u_m\}$
 The diagram shows two diamond-shaped regions representing subspaces of $\mathbb{R}^n$. The top diamond is labeled 'row space' and the bottom diamond is labeled 'null space'. They intersect at a point labeled '0'. The label $\mathbb{R}^n$ is at the bottom right.	 The diagram shows two diamond-shaped regions representing subspaces of $\mathbb{R}^m$. The top diamond is labeled 'column space' and the bottom diamond is labeled 'left null space'. They intersect at a point labeled '0'. The label $\mathbb{R}^m$ is at the bottom right.

There is another way of representing the SVD of A in terms of rank-1 matrices.

$$A = \sigma_1 u_1 v_1^T + \cdots + \sigma_r u_r v_r^T$$

Here, each matrix is of shape $m \times n$ and has rank-1. So we have expressed A as the sum of r unit-rank matrices. One can also use this to write the SVD of A^T :

$$A^T = \sigma_1 v_1 u_1^T + \cdots + \sigma_r v_r u_r^T$$

Two identities follow:

$$\begin{aligned} Av_i &= (\sigma_1 u_1 v_1^T + \cdots + \sigma_r u_r v_r^T) v_i \\ &= \sigma_1 u_1 v_1^T v_i + \cdots + \sigma_r u_r v_r^T v_i \\ &= \sigma_i u_i \end{aligned}$$

Likewise:

$$A^T u_i = \sigma_i v_i$$

To summarize:

$$\begin{array}{l} Av_i = \sigma_i u_i \\ A^T u_i = \sigma_i v_i \end{array}, \quad 1 \leq i \leq r$$

5. Algorithm

Let us perform the SVD of $A = \begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix}$.

Step-1: Compute $A^T A$

$$A^T A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 20 \\ 20 & 80 \end{bmatrix}$$

Step-2: Compute the eigenpairs of $A^T A$

$$\begin{vmatrix} 5 - \lambda & 20 \\ 20 & 80 - \lambda \end{vmatrix} = (5 - \lambda)(80 - \lambda) - 400$$

$$= \lambda^2 - 85\lambda$$

Therefore, the eigenvalues are $\lambda_1 = 85$, $\lambda_2 = 0$. Now for the eigenvectors:

$$\begin{bmatrix} 5 & 20 \\ 20 & 80 \end{bmatrix} - 85 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -80 & 20 \\ 20 & -5 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_2 = 4x_1$$

$(1, 4)$ is one of the eigenvectors of $A^T A$. An eigenvector for $\lambda = 0$ is going to be a vector orthogonal to $(1, 4)$, which is $(-4, 1)$.

$$v_1 = \frac{1}{\sqrt{17}}(1, 4)$$

$$v_2 = \frac{1}{\sqrt{17}}(-4, 1)$$

Step-3: Form the matrices V and Σ

The columns of V are v_1, v_2 :

$$V = \frac{1}{\sqrt{17}} \begin{bmatrix} 1 & -4 \\ 4 & 1 \end{bmatrix}$$

The singular values are the square roots of the eigenvalues of $A^T A$:

$$\begin{aligned} \sigma_1 &= \sqrt{\lambda_1} = \sqrt{85} \\ \sigma_2 &= \sqrt{\lambda_2} = 0 \end{aligned}$$

The singular value matrix:

$$\Sigma = \begin{bmatrix} \sqrt{85} & 0 \\ 0 & 0 \end{bmatrix}$$

Step-4: Get u_i

Recall that $Av_i = \sigma_i u_i$ for $1 \leq i \leq r$ (for all the positive singular values, this relationship is valid)

$$\begin{aligned} Av_1 &= \sigma_1 u_1 \\ u_1 &= \frac{Av_1}{\sigma_1} \end{aligned}$$

Plug in the values:

$$\begin{aligned} u_1 &= \frac{1}{\sqrt{85}} \frac{1}{\sqrt{17}} \begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} \\ &= \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \end{aligned}$$

We now have one column of U . The other column of U is orthogonal to

u_1 . One possibility for $u_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. With that U becomes:

$$U = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

Finally, the SVD of A :

$$\begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{85} & 0 \\ 0 & 0 \end{bmatrix} \cdot \frac{1}{\sqrt{17}} \begin{bmatrix} 1 & 4 \\ -4 & 1 \end{bmatrix}$$

Uses of SVD

- Low-rank approximation of A : Truncate the SVD after the first k terms. If

$$A = \sigma_1 u_1 v_1^T + \dots + \sigma_r u_r v_r^T$$

The best rank- k matrix that approximates A is given by:

$$\hat{A} = \sigma_1 u_1 v_1^T + \dots + \sigma_k u_k v_k^T$$

- Image compression: An image is a matrix. So by using a low-rank approximation that SVD gives, one can compress a matrix.
- SVD allows us to compute the pseudo-inverse of a matrix. The pseudo-inverse is defined for any matrix.

$$A^\dagger = V \Sigma^\dagger U^T$$

This is used to compute what is called the least-squares solution in regression. (week-3)

$$\hat{x} = (A^T A)^{-1} A^T b$$

$$\hat{x} = A^\dagger b$$

- PCA is SVD
 - PCA is SVD done on a centered data-matrix. Then the left singular vectors directly give you the PCs.
 - PCA: The eigenvectors of the covariance matrix are the principal components. $\frac{1}{n}XX^T \rightarrow w_1, \dots, w_d$
 - SVD: The left singular vectors of X are the principal components.

6. Definiteness and Quadratic Forms

For some motivation, the idea of a quadratic form makes an appearance in optimization. What is quadratic form? It is a polynomial in n variables in which each term has degree 2. For example:

$$3x_1^2$$

$$x_1^2 - 2x_1x_2 + x_2^2$$

$$x_1^2 - x_2^2 + x_3^2 + 2x_1x_2 - 5x_2x_3 + 17x_3x_1$$

What do quadratic forms have to do with matrices? Every quadratic form is associated with a matrix.

$$A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$x_1^2 - 2x_1x_2 + x_2^2 = x^T Ax$$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 - x_2 \\ -x_1 + x_2 \end{bmatrix}$$

$$= x_1(x_1 - x_2) + x_2(-x_1 + x_2)$$

$$= x_1^2 - 2x_1x_2 + x_2^2$$

The same quadratic form can be associated with another matrix:

$$B = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Notice that $x^T Bx = x^T Ax$.

Lemma: Every quadratic form is associated with a unique symmetric matrix.

If you are given a matrix, how do you convert it to the corresponding quadratic form?

$$A = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The quadratic form:

$$f(x) = 1 \cdot x_1^2 + 7 \cdot x_1x_2 + 3 \cdot x_2^2$$

Another example:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -3 & 4 \\ -1 & 4 & 5 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$f(x) = x_1^2 - 3x_2^2 + 5x_3^2 + 4x_1x_2 - 2x_1x_3 + 8x_2x_3$$

$$\begin{bmatrix} & x_1 & x_2 & x_3 \\ x_1 & 1 & 2 & -1 \\ x_2 & 2 & -3 & 4 \\ x_3 & -1 & 4 & 5 \end{bmatrix}$$

If you are given a quadratic form, how do you find out the corresponding symmetric matrix?

For example:

$$f(x) = x_1^2 - 3x_2^2 + 5x_3^2 + 4x_1x_2 - 2x_1x_3 + 8x_2x_3$$

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -3 & 4 \\ -1 & 4 & 5 \end{bmatrix}$$

A non-symmetric example:

$$B = \begin{bmatrix} 1 & 3 & -2 \\ 1 & -3 & 2 \\ 0 & 6 & 5 \end{bmatrix}$$

In general, a quadratic form in n variables is expressed as:

$$f(x) = x^T A x$$

One can also express this as:

$$f(x) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

- Each term has degree 2.
- If you set $i = j$, you get the diagonal element A_{ii} and also the pure square term x_i^2 .
- As for the off-diagonal symmetric pair, $x_i x_j$, its coefficient will be $A_{ij} + A_{ji}$.

Where does this appear?

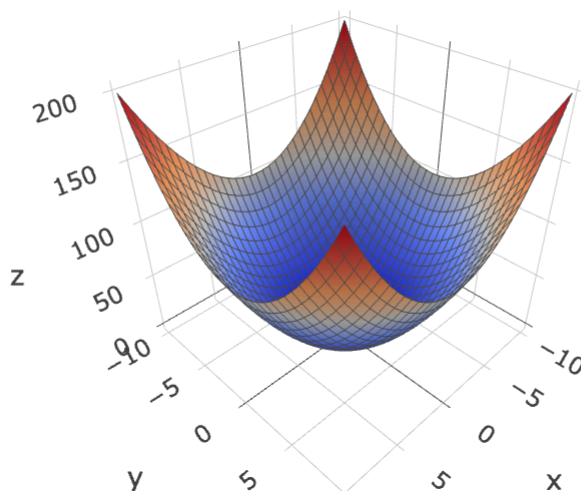
$$f(x) \approx f(x_0) + \nabla f(x_0)^T (x - x_0) + \frac{1}{2!} (x - x_0)^T H (x - x_0)$$

where $H = \nabla^2 f$.

A quadratic form is closely tied to bowl-shaped surfaces.

Examples:

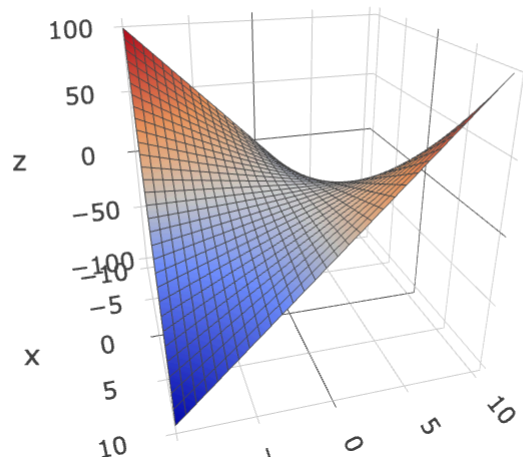
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$f(x, y) = x^2 + y^2$$

$$A = \begin{bmatrix} 0 & 0.5 \\ 0.5 & 0 \end{bmatrix}$$

$$f(x, y) = xy$$



We want to take a closer look at bowl-shaped quadratic forms because they are useful in the study of optimization. When is a quadratic form a bowl-shaped surface? When the associated symmetric matrix is positive definite, a concept we define below.

Positive Definite Matrices

Definition: A positive definite matrix is a symmetric matrix A such that $x^T A x > 0$ whenever $x \neq 0$.

How do we find out if a matrix is positive definite?

Lemma-1: A symmetric matrix A is positive definite if and only if A has positive eigenvalues.

One direction: If A is p.d., then its eigenvalues are positive.

If (λ, v) is an eigenpair of a positive definite matrix:

$$\begin{aligned} Av &= \lambda v \\ v^T Av &= \lambda v^T v \end{aligned}$$

Since A is p.d. $v^T Av$ is also positive. This means that $\lambda(v^T v)$ is also positive. Therefore, λ is positive.

Lemma-2: A symmetric matrix A is positive definite if and only if A can be expressed as $B^T B$ (BB^T) for some invertible matrix B .

As an application of lemma-1, for example, consider:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 8 \end{bmatrix}$$

$$(1 - \lambda)(8 - \lambda) - 4 = 0$$

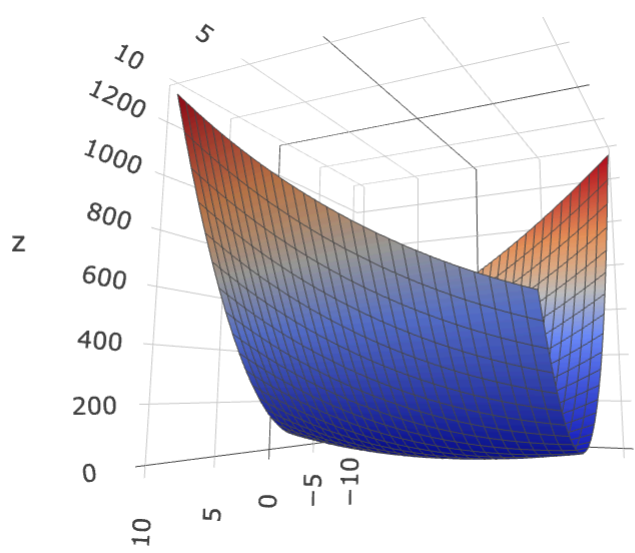
$$\lambda^2 - 9\lambda + 4 = 0$$

$$\lambda = \frac{9 \pm \sqrt{81 - 16}}{2} = \frac{9 \pm \sqrt{65}}{2}$$

$$\lambda_1, \lambda_2 > 0$$

Therefore, we can conclude that A is p.d. The corresponding quadratic form is:

$$\begin{bmatrix} 1 & 2 \\ 2 & 8 \end{bmatrix} \rightarrow x^2 + 4xy + 8y^2$$



Definiteness	Definition	Condition
Positive definite	$x^T A x > 0$ for $x \neq 0$	Positive eigenvalues

Positive semi-definite	$x^T Ax \geq 0$	Non-negative eigenvalues
Negative definite	$x^T Ax < 0$ for $x \neq 0$	Negative eigenvalues
Negative semi-definite	$x^T Ax \leq 0$	Non-positive eigenvalues
Indefinite	None of the above some are positive, some are negative	None of the above some are positive, some are negative

In the lectures:

$$f(x,y) = ax^2 + 2bxy + cy^2$$

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

For A to be p.d, we need the eigenvalues to be positive:

$$(a-\lambda)(c-\lambda) - b^2$$

$$\lambda^2 - (a+c)\lambda + ac - b^2 = 0$$

This is what we want:

$$\lambda_1 > 0, \lambda_2 > 0$$

This is equivalent to the following:

$$\begin{aligned} \lambda_1 + \lambda_2 &> 0 \\ \lambda_1 \lambda_2 &> 0 \end{aligned}$$

Which is equivalent to:

$$a + c > 0$$

$$ac > b^2$$

Which is equivalent to:

$$a > 0$$

$$ac > b^2$$